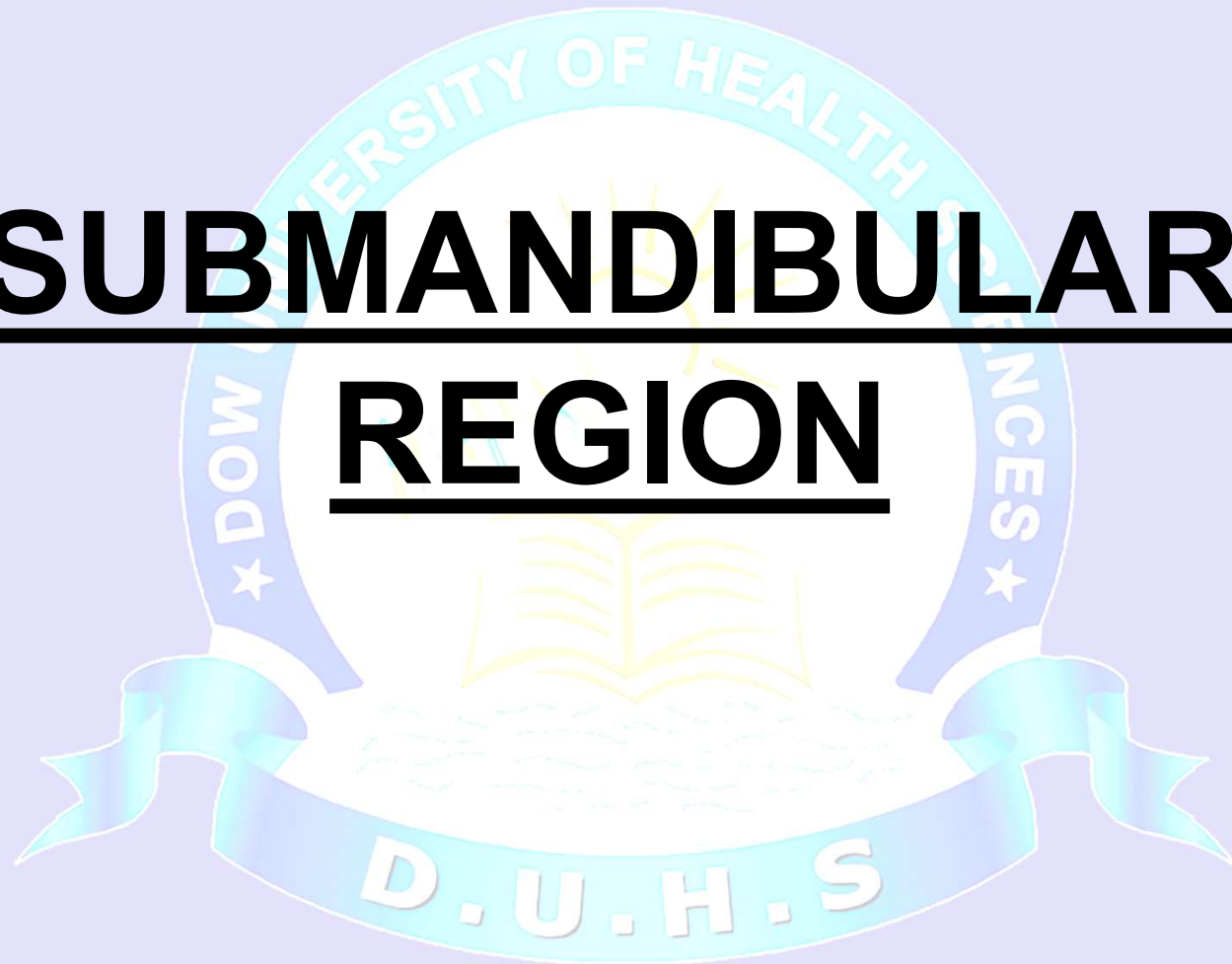


SUBMANDIBULAR

REGION



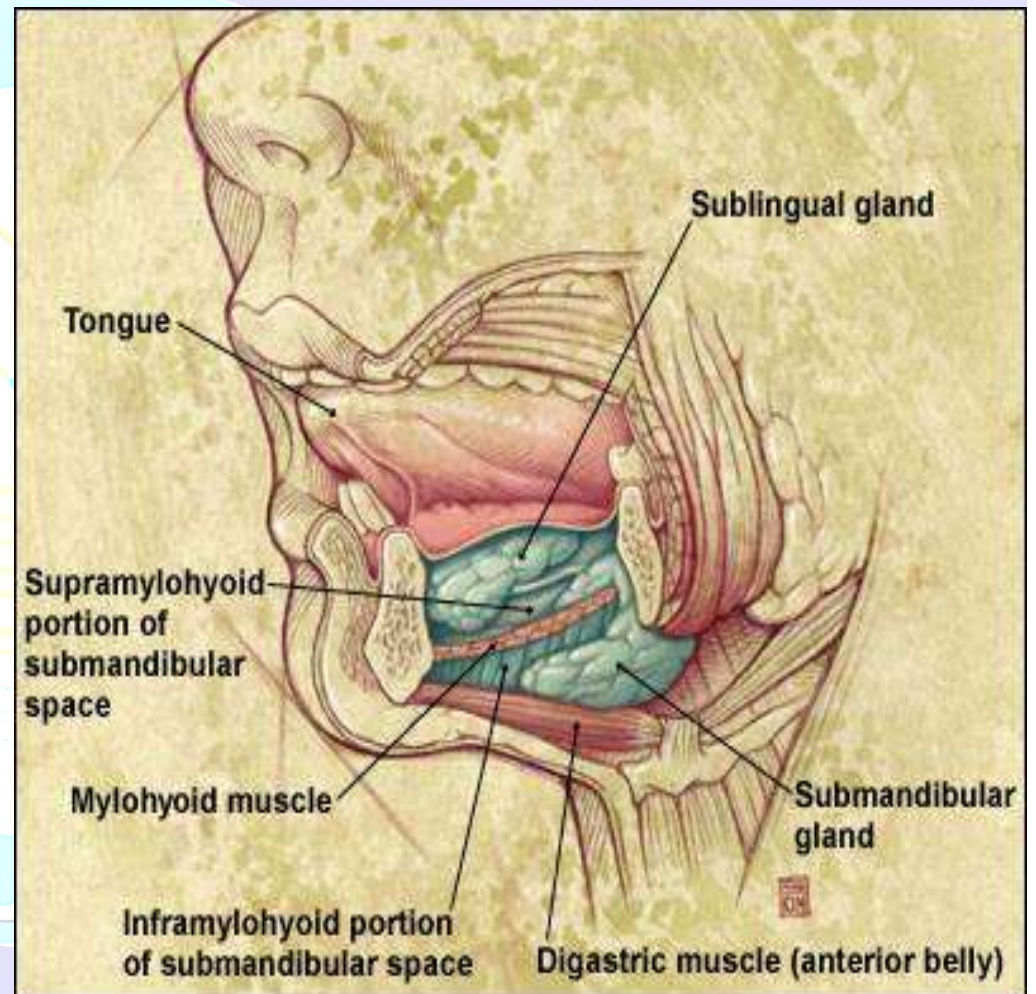
LEARNING OBJECTIVES

At the end of the lecture, the student should be able to :

- Explain the **submandibular region**.
- Describe the **anterior triangle of the neck**.
- Tell the **Suprahyoid muscles**.
- Describe the **submandibular gland**.
- Describe the **sublingual gland**.
- What is **submandibular ganglion**?

SUBMANDIBULAR REGION

- Region between the mandible and hyoid bone contains **submandibular glands, sublingual glands, Suprahyoid muscles, submandibular ganglion, and lingual artery.**



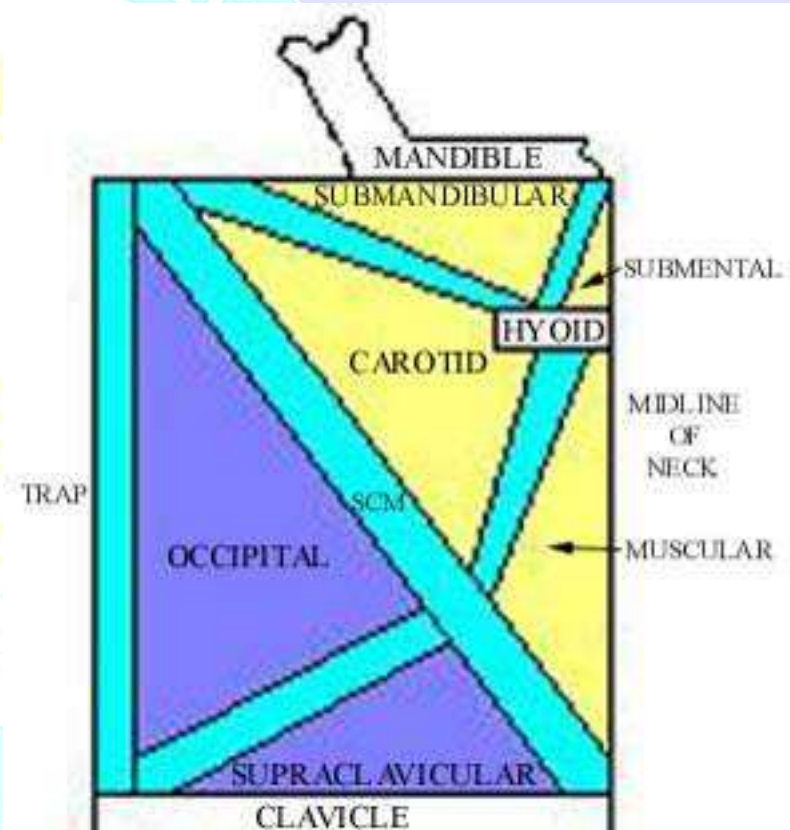
ANTERIOR TRIANGLE OF THE NECK

- Outlined by:

- Anterior border of sternocleidomastoid muscle laterally,
- Inferior border of the mandible superiorly,
- And midline of the neck medially.

- Further subdivided into:

- Submandibular triangle
- Submental triangle
- muscular triangle
- carotid triangle



ANTERIOR TRIANGLE OF THE NECK

Submandibular triangle

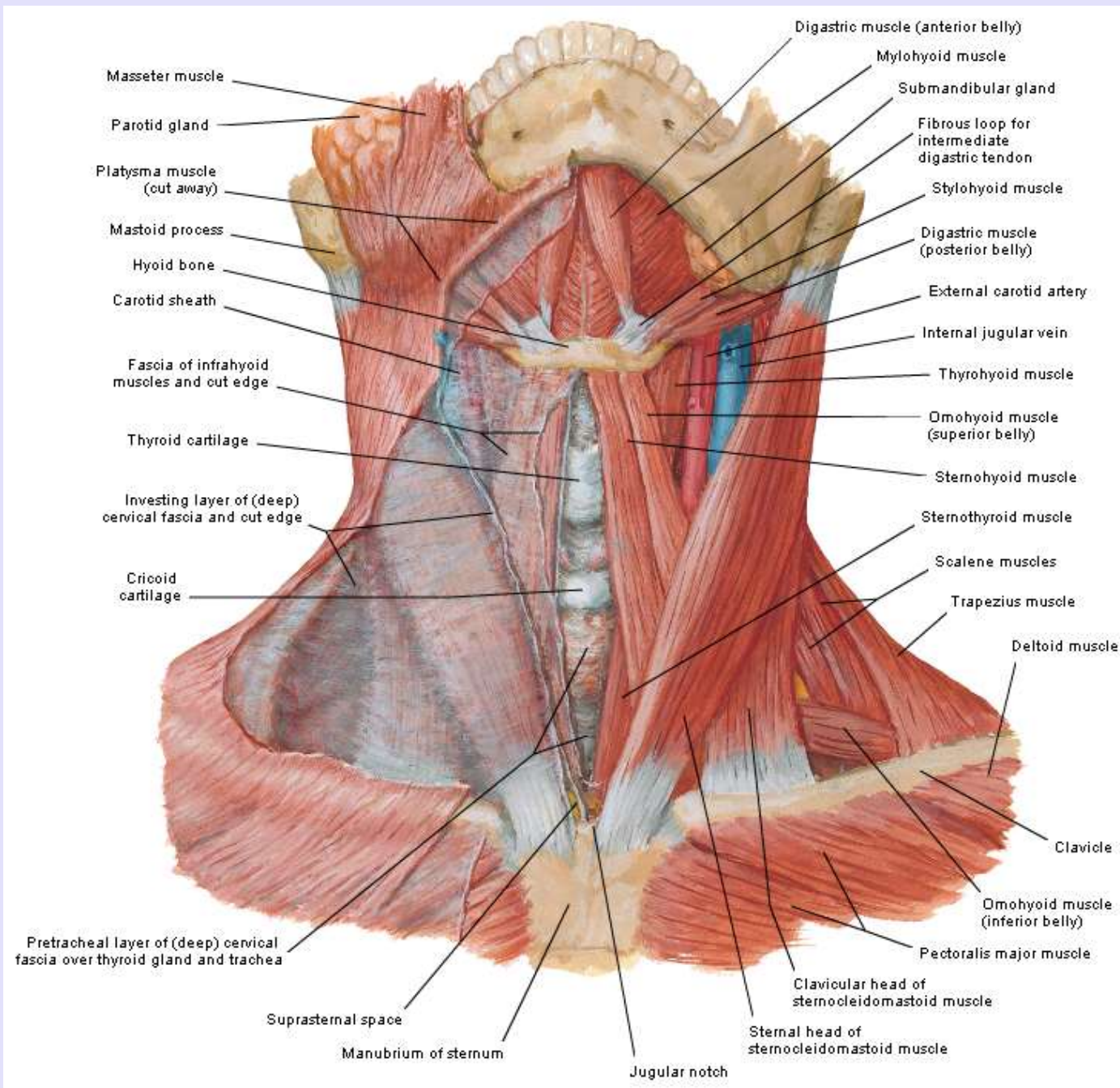
– Outlined by:

- Inferior border of the mandible superiorly and
- Anterior and posterior bellies of the digastric muscle inferiorly.

Submental triangle

– Outlined by:

- Hyoid bone inferiorly,
- Anterior belly of digastric muscle laterally, and the midline.



ANTERIOR TRIANGLE OF THE NECK

Muscular Triangle

- Outlined by:

- hyoid bone superiorly,
- superior belly of omohyoid muscle,
- anterior border of the sternocleidomastoid muscle laterally, and the midline;

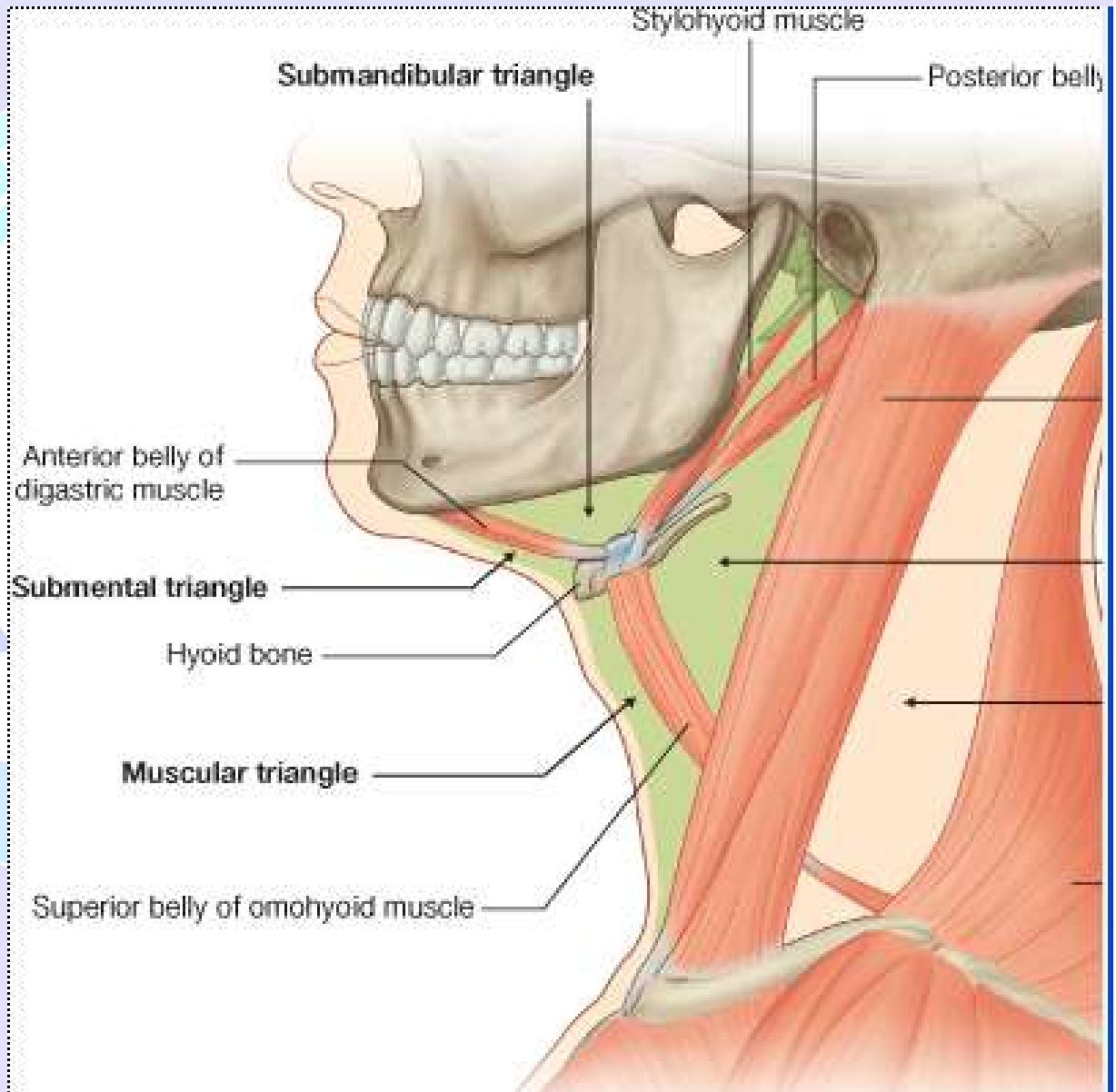
Carotid Triangle

- Outlined by:

- superior belly of omohyoid muscle anteroinferiorly,
- stylohyoid muscle and posterior belly of digastric superiorly
- anterior border of sternocleidomastoid muscle posteriorly

Borders and subdivisions of the anterior triangle of neck.

Muscles Superior To Hyoid Are classified as **Suprahyoid Muscles And Include the Stylohyoid, Digastric, Mylohyoid, And Geniohyoid.**

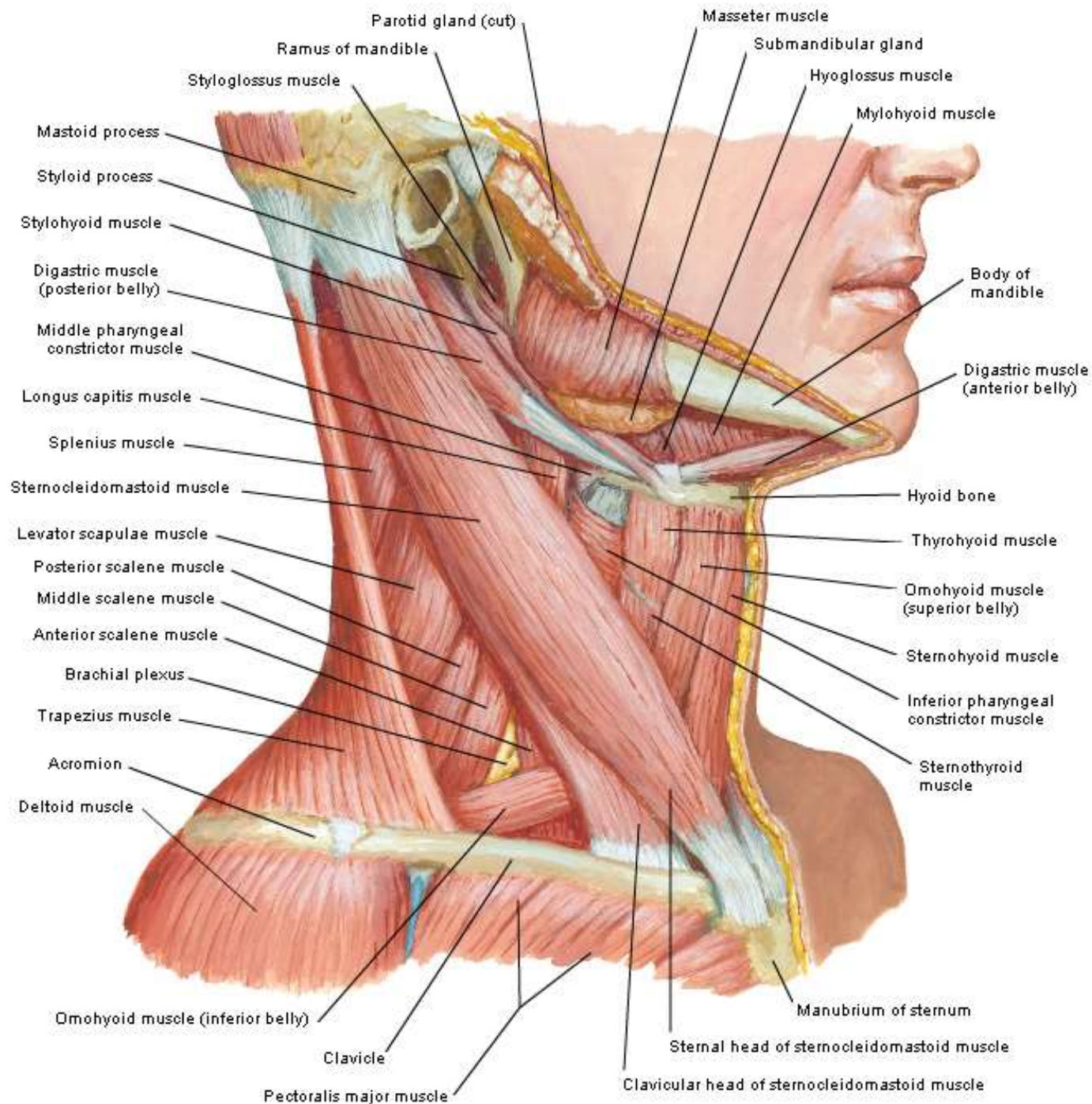


Muscle	Origin	Insertion	Innervation	Function
Mylohyoid	Base of styloid process	Lateral area of body of hyoid bone	Facial nerve [VII]	Pulls hyo posterio
Digastric anterior belly	Digastric fossa on lower inside of mandible	Attachment of tendon between two bellies to body of hyoid bone	Mylohyoid nerve from inferior alveolar branch of mandibular nerve [V ₃]	Opens m
Digastric posterior belly	Mastoid notch on medial side of mastoid process of temporal bone		Facial nerve [VII]	
Mylohyoid	Mylohyoid line on mandible	Body of hyoid bone and fibers from muscle on opposite side	Mylohyoid nerve from inferior alveolar branch of mandibular nerve [V ₃]	Support & of mouth;
Geniohyoid	Inferior mental spine on inner surface of mandible	Anterior surface of body of hyoid bone	Branch from anterior ramus of C1 (carried along the hypoglossal nerve [XII])	Fixed ma pulls hyo fixed hyo mandible ward

SUPRAHYOID MUSCLES

Suprahyoid muscles are in submental and submandibular triangles.

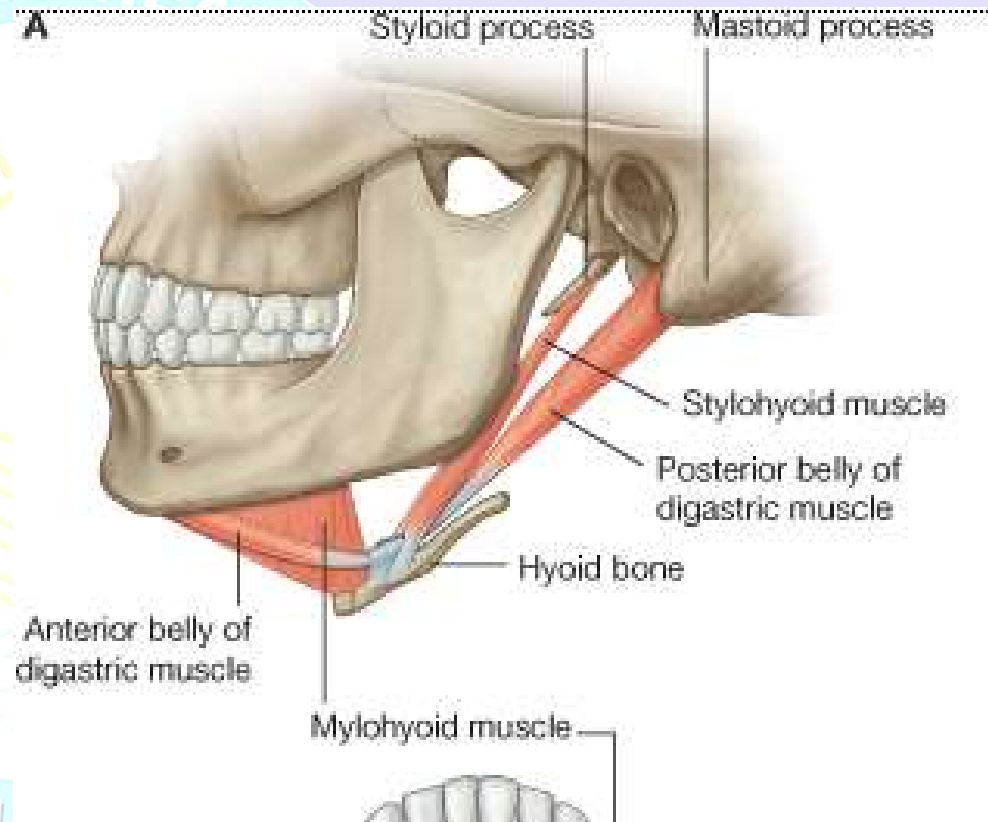
- Pass in a superior direction from hyoid bone to skull or mandible and raise the hyoid, as occurs during swallowing.
- **Stylohyoid** muscle arises from base of styloid process and passes anteroinferiorly to attach to lateral area of body of hyoid bone.
- During swallowing it pulls hyoid bone posterosuperiorly and it is innervated by facial nerve .



Digastric Muscle

Has two bellies connected by a tendon, which attaches to the body of the hyoid bone.

- **Posterior belly** arises from mastoid notch on medial side of the mastoid process of temporal bone;
- **Anterior belly** arises from digastric fossa on the lower inside of mandible.
- Tendon between two bellies, which is attached to body of hyoid bone, is the point of insertion of both bellies.

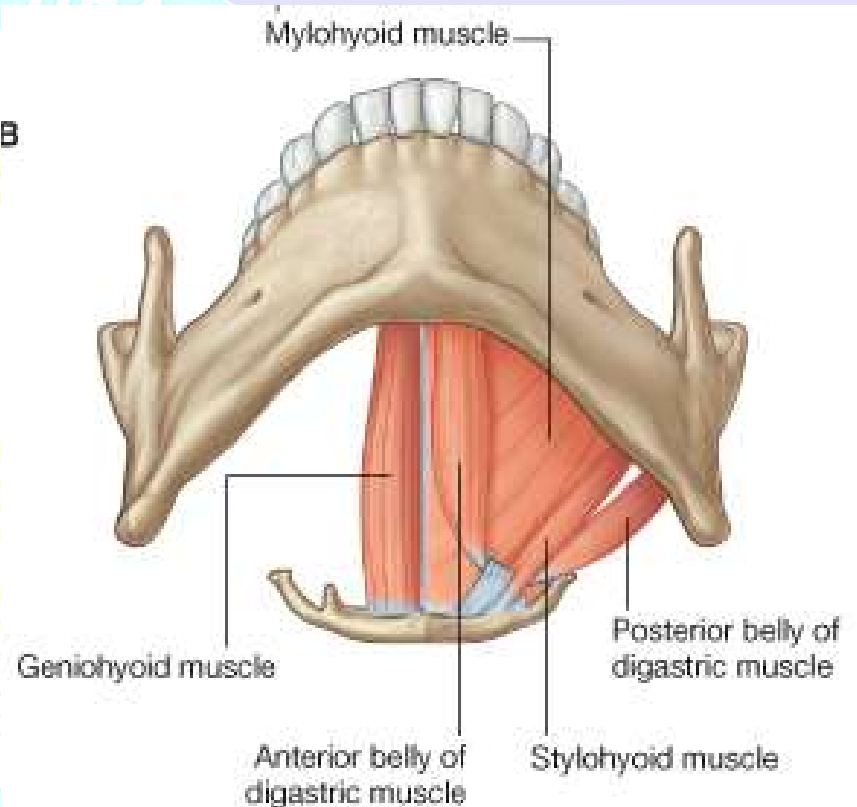


Digastric Muscle

- Has multiple actions depending on which bone is fixed:
- When mandible is fixed, digastric raises hyoid bone;
- When hyoid bone is fixed, digastric opens the mouth by lowering mandible.
- Innervation of muscle is from two different cranial nerves.
- Innervation of posterior belly of digastric muscle is by facial nerve whereas anterior belly of muscle is innervated by trigeminal nerve.

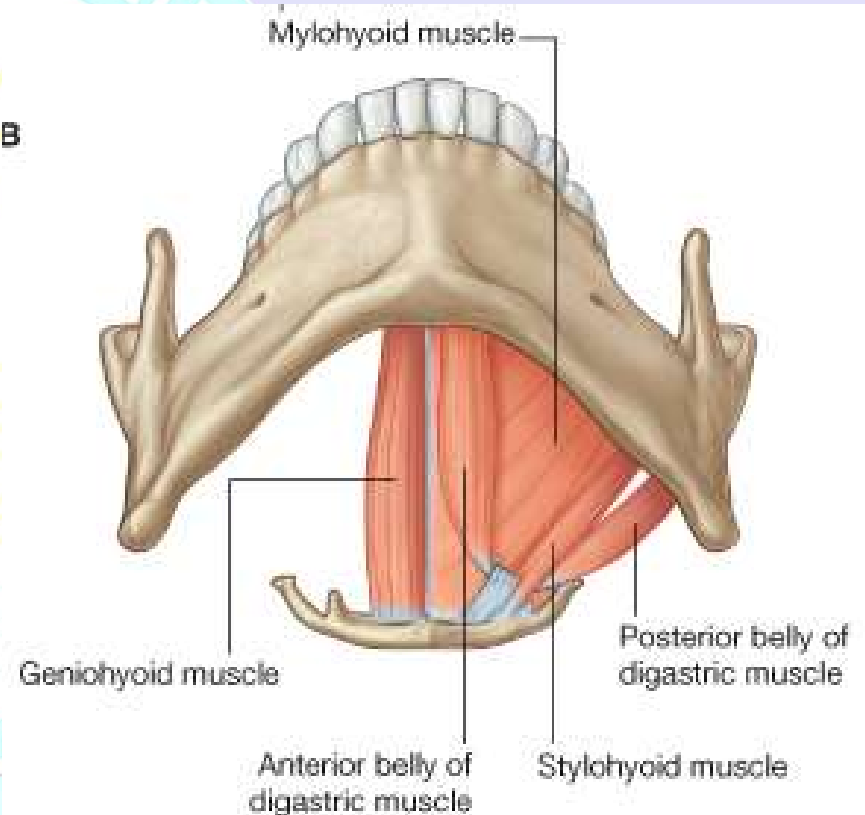
Mylohyoid Muscle

- It is superior to anterior belly of digastric and, with its partner from opposite side, forms floor of mouth .
- Originates from mylohyoid line on the mandible and inserts into hyoid bone and also blends with mylohyoid muscle from the opposite side.
- Supports and elevates the floor of mouth and elevates hyoid bone.
- Innervated by the trigeminal nerve.



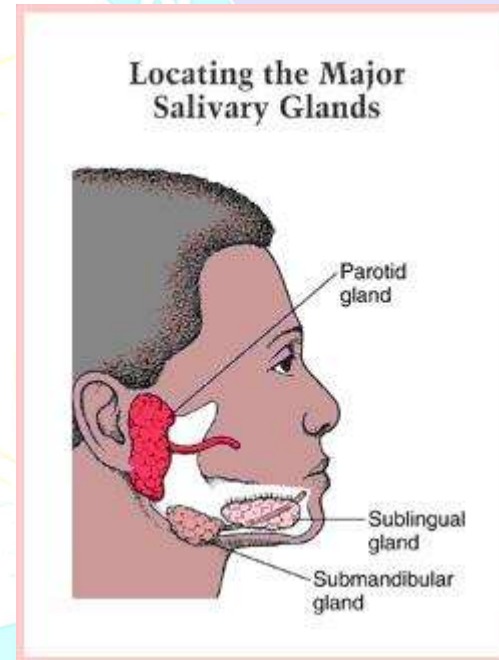
THE GENIOHYOID

- A narrow muscle, it is superior to the medial part of each mylohyoid muscle.
- Muscles from each side are next to each other in midline.
- Arises from inferior mental spine of mandible and passes backward and downward to insert on body of hyoid bone.
- Has two functions depending on which bone is fixed:
 - Fixation of mandible elevates and pulls hyoid bone forward;
 - Fixation of hyoid bone pulls mandible downward and inward.
- Innervated by a branch from anterior ramus of c1 carried along hypoglossal nerve .



Salivary Glands

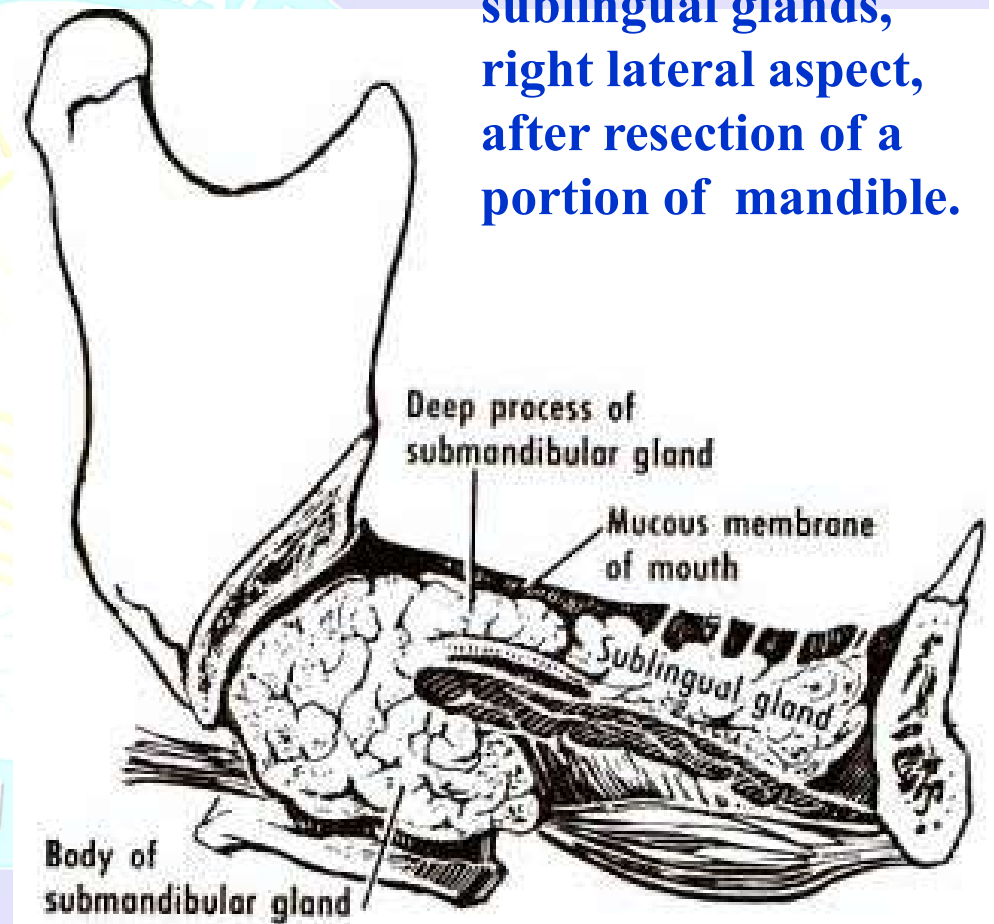
- Large, paired salivary glands are the parotid, submandibular, and sublingual glands.



SUBMANDIBULAR GLAND

- is usually just palpable.
- It has a larger superficial part (body) and a smaller deep process.
- two parts are continuous with each other, forming a "U" shape around posterior border of mylohyoid muscle.
- body of gland is in and inferior to digastric triangle and also partly under cover of mandible.

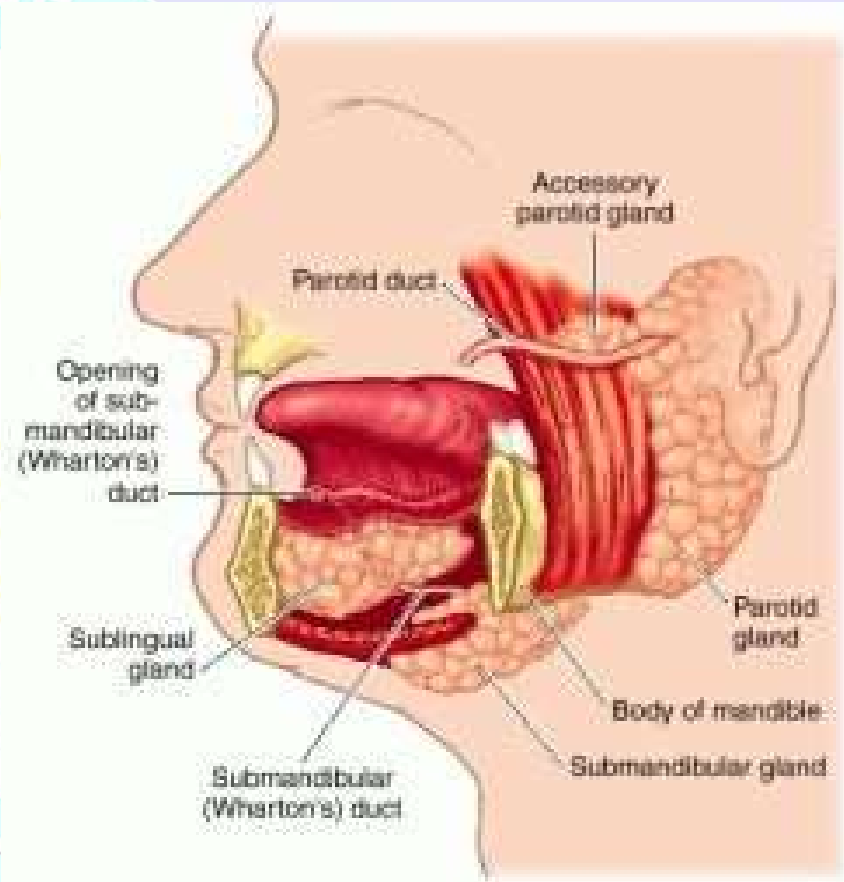
submandibular and sublingual glands, right lateral aspect, after resection of a portion of mandible.



SUBMANDIBULAR GLAND

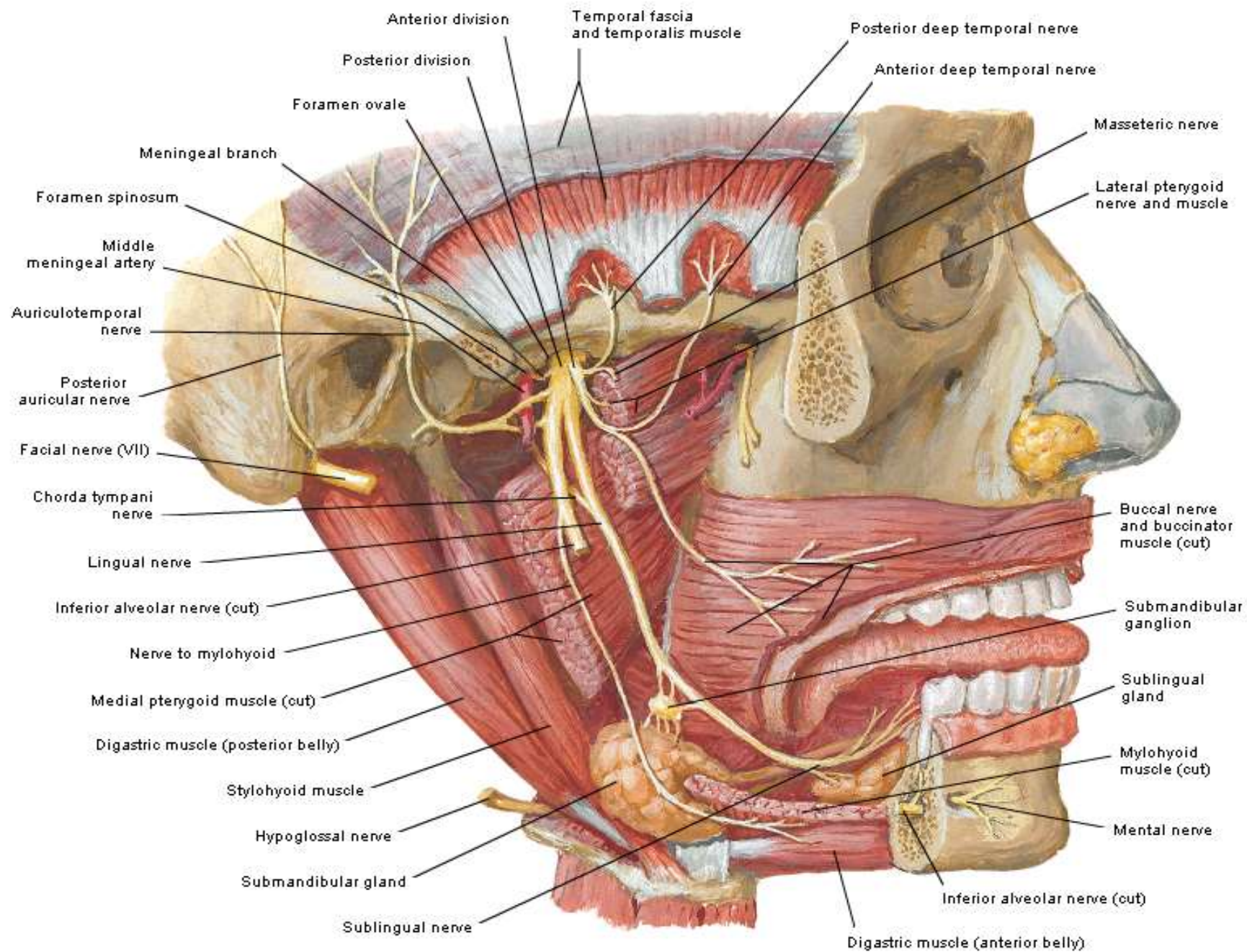
Has three surfaces:

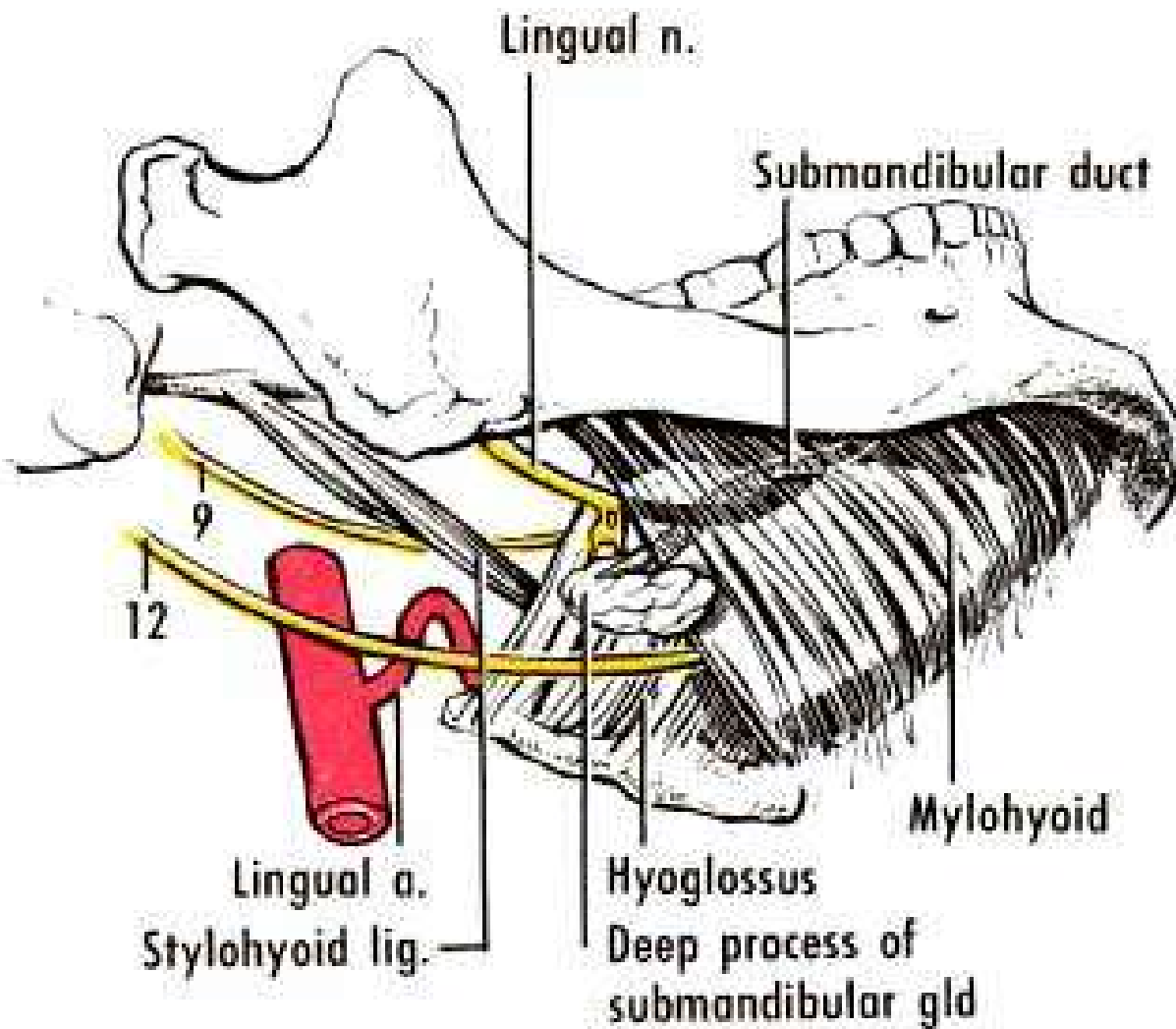
- **Inferior (covered by skin and platysma), lateral (related to the medial surface of the mandible), and medial (related to the mylohyoid, hyoglossus, and digastric muscles).**
- **Deep process lies between mylohyoid and hyoglossus muscles the submandibular duct, which is crossed by the lingual nerve, runs anterior from this process**
- **Submandibular duct opens by one to three orifices into the oral cavity on the sublingual papilla, at the side of the frenulum linguae.**



SUBMANDIBULAR GLAND

- Branches of the duct can be examined radiographically after injection of a radio-opaque medium (sialography).
- Gland is supplied by postganglionic parasympathetic, secretomotor fibers derived from the submandibular ganglion.
- Preganglionic fibers leave the skull with chorda tympani (a branch of facial nerve) and then join lingual nerve on their way to submandibular ganglion.

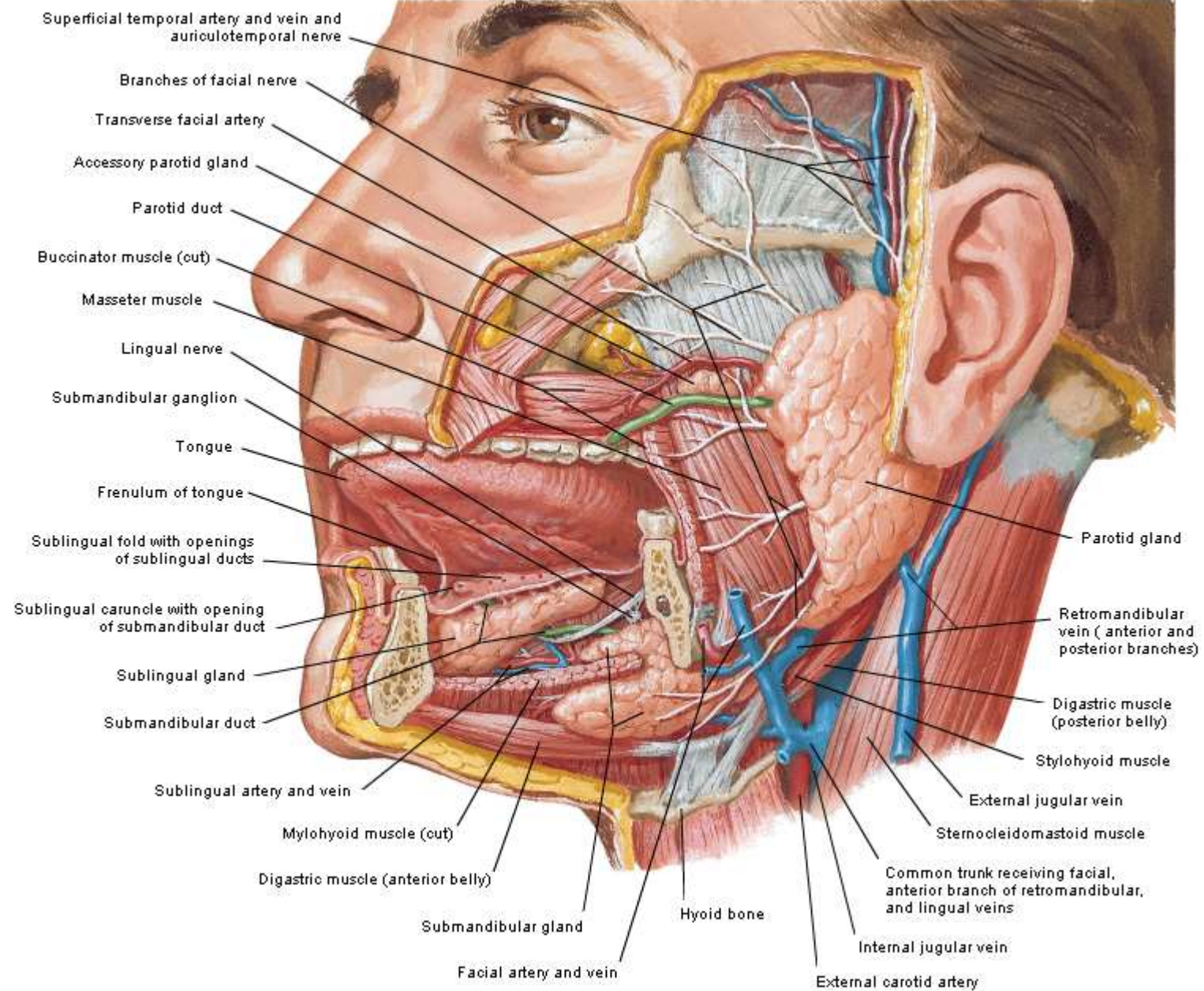


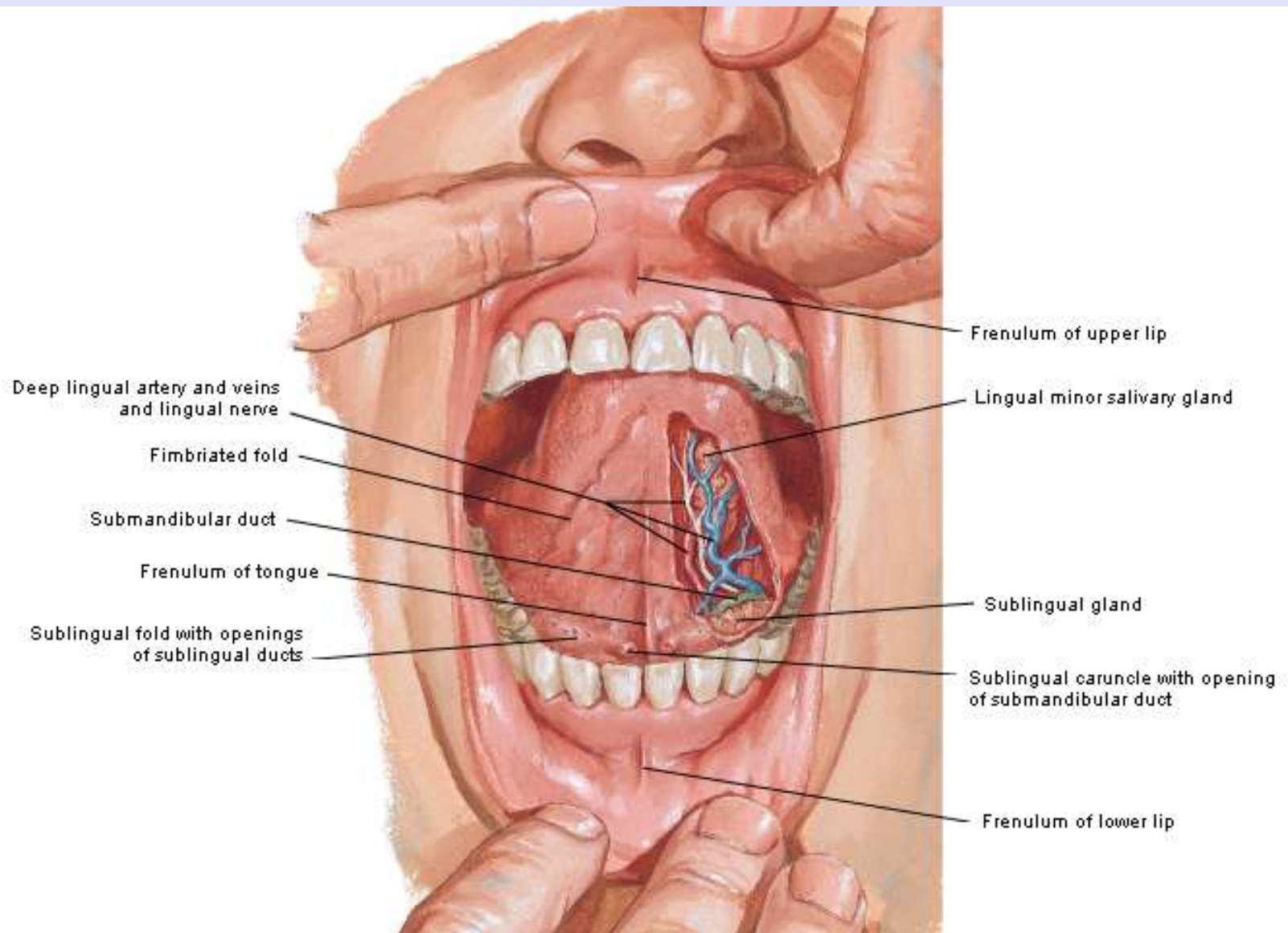


Relations of mylohyoid and hyoglossus muscles. The lingual nerve, the deep process of the submandibular gland and the submandibular duct, and the hypoglossal nerve pass deep to the posterior border of the mylohyoid. The glossopharyngeal nerve, the stylohyoid ligament, and the lingual artery pass deep to the posterior border of the hyoglossus.

SUBLINGUAL GLAND

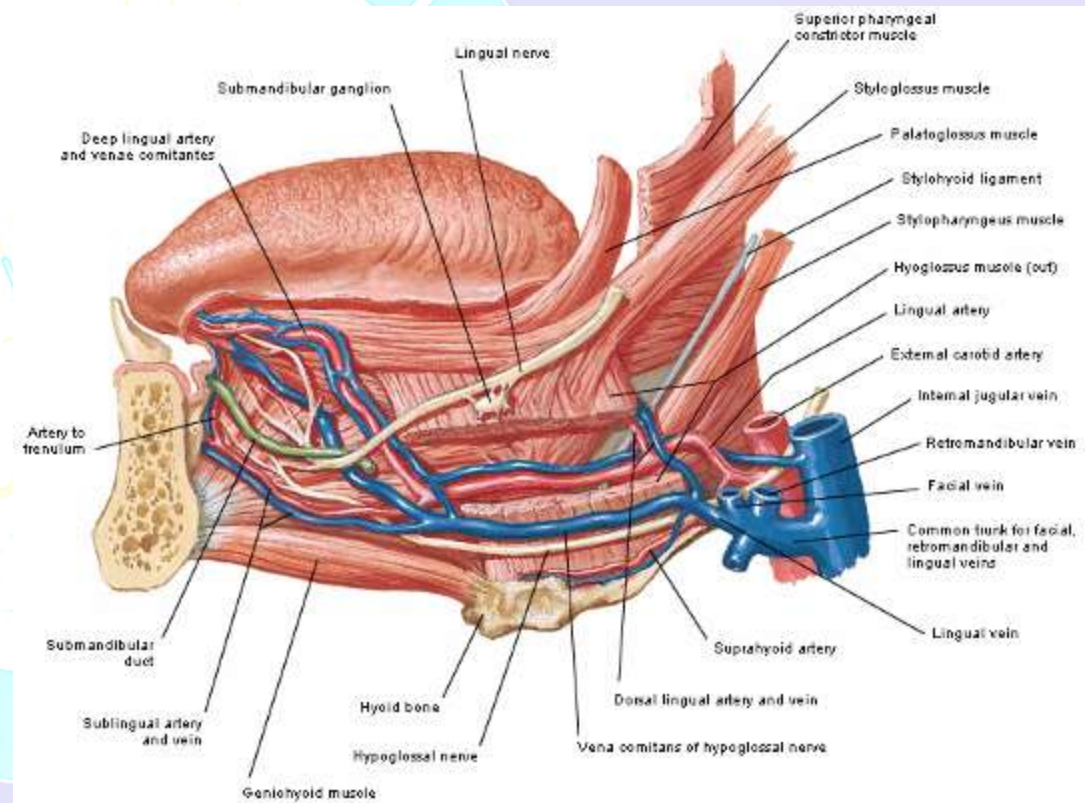
- inferior to mucosa of floor of mouth and is anterior to deep process of submandibular gland.
- Most of the 10 to 30 sublingual ducts open separately into the floor of oral cavity.
- innervation of the gland from the submandibular ganglion is similar to that of the submandibular gland.





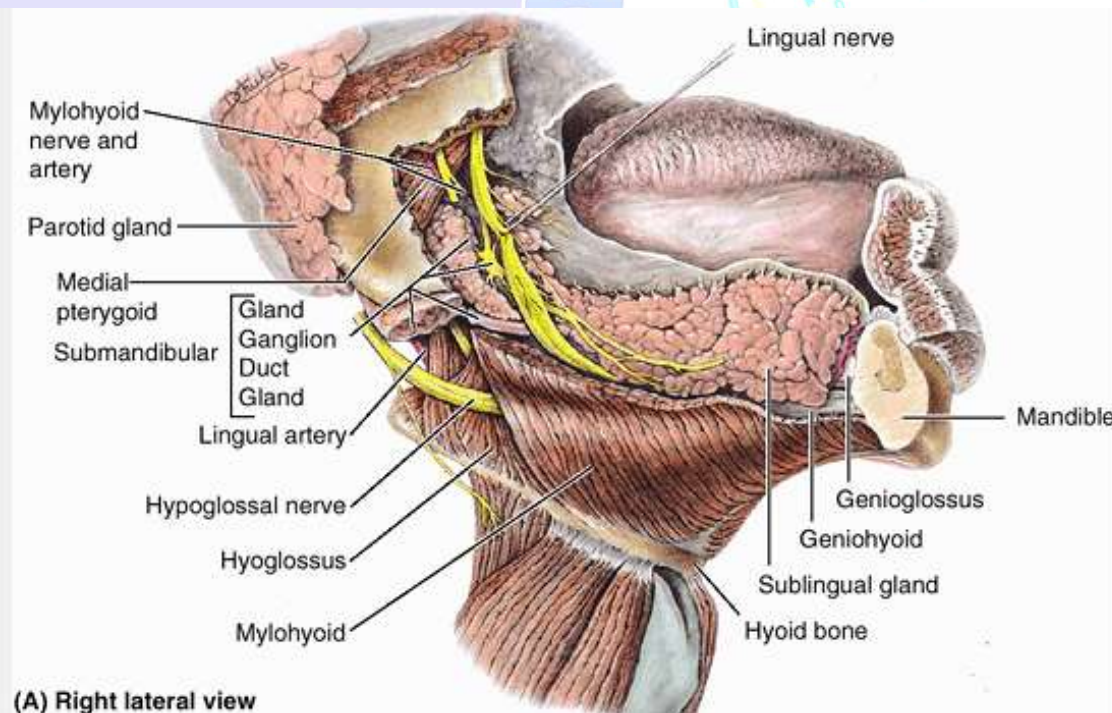
SUBMANDIBULAR GLAND

- on lateral surface of hyoglossus muscle, medial to the mylohyoid muscle, superior to the submandibular duct and hypoglossal nerve, and inferior to the lingual nerve, from which it is suspended by several branches.
- Preganglionic parasympathetic fibers derived from chorda tympani travel in lingual nerve and synapse in submandibular ganglion.

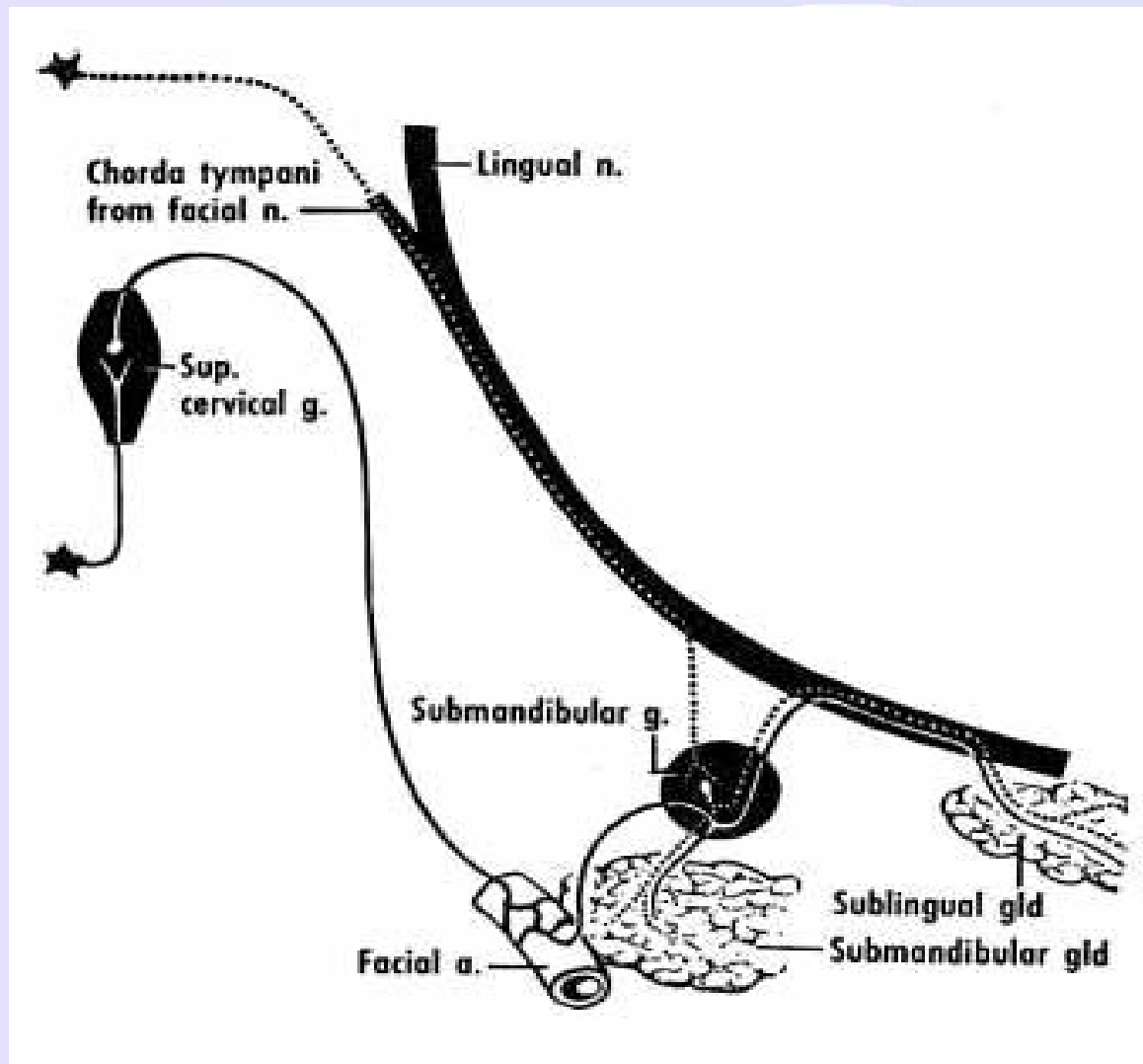


SUBMANDIBULAR GANGLION

- Some of the postganglionic secretory fibers enter the submandibular gland; others, by entering the lingual nerve, reach the sublingual gland.
- Postganglionic sympathetic fibers (from the superior cervical ganglion) pass through the submandibular ganglion and are distributed with the parasympathetic fibers.



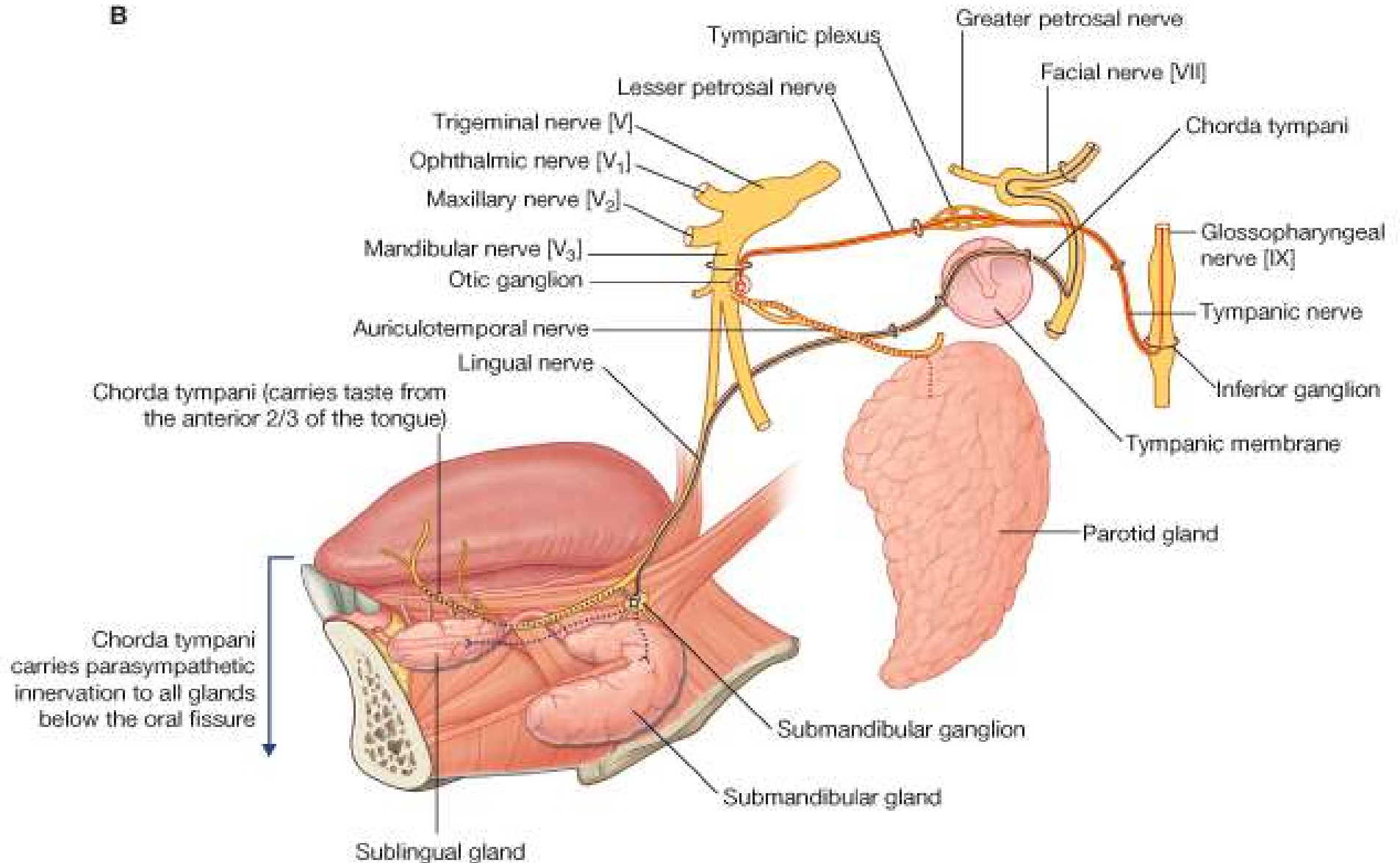
SUBMANDIBULAR GANGLION



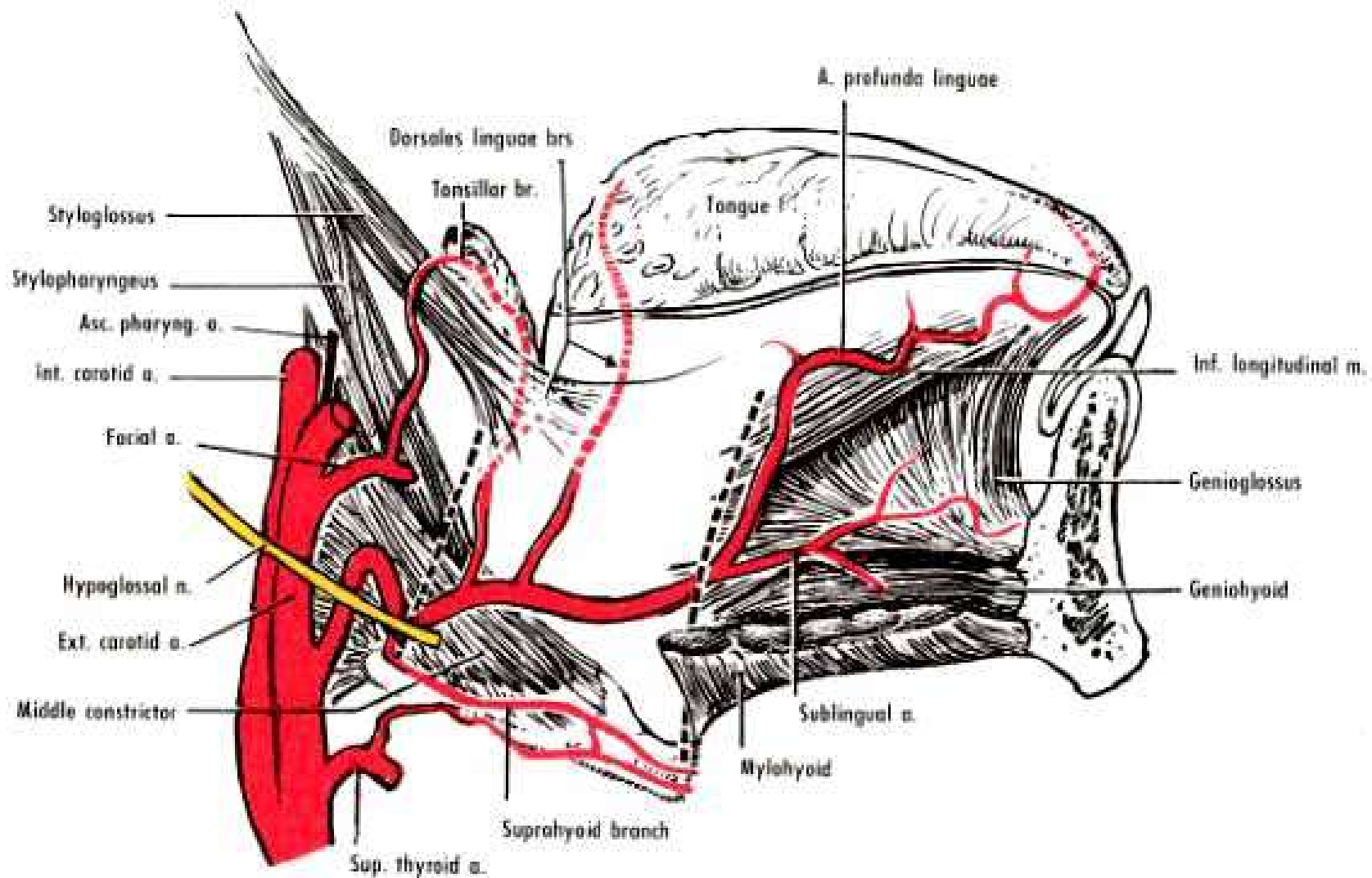
The submandibular ganglion and its connections, lateral aspect. The dotted lines represent parasympathetic fibers. Sympathetic fibers are shown by continuous lines.

CHORDA TYMPANI AND NERVE SUPPLY OF SUB MANDIBULAR REGION

B



The lingual artery. The interrupted lines indicate the borders of the resected hyoglossus muscle



LINGUAL ARTERY

- Lingual artery, a branch of the external carotid artery.
- The dorsal lingual arteries provide the blood supply to the root of the tongue and a branch to the palatine tonsil. ★
- The deep lingual arteries supply the body of the tongue. The sublingual arteries provide the blood supply to the floor of the mouth, including the sublingual glands.



THANK YOU